Fiscal Year:	FY 2007	**Task Last Updated:**	FY 10/03/2007
PI Name:	Butel, Janet S. Ph.D.		
Project Title:	Biology of Virus Infections: Radiation and Immunity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030-3411	**Congressional District:**	25
Comments:	Address updated 9/2008		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	12/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Conner, Margaret (Baylor College of Medicine) Ling, Paul (Baylor College of Medicine) Zhang, Shaojie (Baylor College of Medicine)		
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Task Description:	Space flight has been found to affect immune responses, and alterations in a normal immune response often have a major impact on the host's ability to control infections. A question being explored is whether infectious diseases pose a medical risk to the success of long-duration space journeys. All humans are infected for life with latent and persistent viruses, and it is well-known that suppression of the immune system allows latent viruses to reactivate and multiply, which may cause disease in the person undergoing reactivation or in contacts to whom the virus is transmitted. The general hypothesis is that conditions of long-duration space flight, including radiation, stress, isolation, containment, and sleep deprivation, will alter human immune responses, leading to reactivation of latent viruses, increased viral infections and viral disease, and possible development of malignancies. We focused on reactivation and shedding of human herpesvirus EBV and human polyomaviruses, agents known to establish persistent infections and to undergo reactivation and cause disease, including cancer, when the host immune system is compromised. Animal models were used to study radiation effects on viral infections and host responses to those infections. The specific aims of the project were: (1) To determine the effects of space radiation and hind limb unloading (HLU) on host control of virus infections and virus-induced cancers in a mouse model; and (2) To characterize direct effects of radiation on viruses and virus-infected cells. Key findings of the project were from the mouse polyoma virus space radiation model we developed. A quantitative real-time polymerase chain reaction (RQ-PCR) assay was developed to measure murine polyoma virus (MuPyV) genome copies in infected animal tissues. This model showed that gamma-irradiation leads to immunosuppression, delayed clearance of primary virus infection, and reactivated latent viral infections. Both single high-dose and multiple low doses of gamma-irradiation cause virus reactivation. HLU, which simulates aspects of weightlessness, was applied to the mouse model. HLU results in loss of control of virus infection in a tissue-specific manner. Another finding was the identification of a herpesvirus EBV latent protein that might modulate the development of EBV-associated disease. In collaboration with Dr. G. Sonnenfeld we carried out measurements of virus reactivations in volunteers participating in the NASA-sponsored Bed Rest Study, a model for the effects of space flight conditions on human physiology.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This NSBRI-funded research addressed space flight-induced alterations in the immune system and their effects on host control of microbial infections. We have shown in several ground-based human models of space flight that even modest depressions in immune function correlate with virus reactivation and shedding. Using a mouse model, we have evidence of radiation-induced reactivation of viral infections and depression of host immune function. New information from these studies are applicable to the analysis of risks that may affect long-duration space travel. The knowledge gained from studies of virus infections in these test models will be applicable to earth-bound individuals at risk of suffering similar virus reactivations and serious, sometimes life-threatening, consequences due to immunosuppression following organ transplantation or cancer chemotherapy and during pregnancy, old age, and AIDS. Studies of the role of herpesvirus EBV latent genes in human disease may lead to the development of novel antiviral drugs of benefit to individuals on earth experiencing virus reactivations.
Task Progress:	Key findings of the project this year were from the mouse polyoma virus space radiation model we developed. This model showed that gamma-irradiation leads to immunosuppression, delayed clearance of primary virus infection, and reactivated latent viral infections. Both single high-dose and multiple low doses of gamma-irradiation cause virus reactivation. Hind limb unloading (HLU), that simulates aspects of weightlessness, was applied to the mouse model and resulted in loss of control of virus infection in a tissue-specific manner. These findings show the deleterious effect of multiple low-dose exposures to irradiation on host control of infections and the added adverse effect of HLU. Significant efforts were made this past reporting period to develop conditions for microarray analyses of immune response gene expression in mice under conditions of virus infection and/or irradiation. This approach will allow an understanding of the mechanisms underlying deleterious effects of space flight conditions on host control of infections.
Bibliography Type:	Description: (Last Updated: 04/25/2011)
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